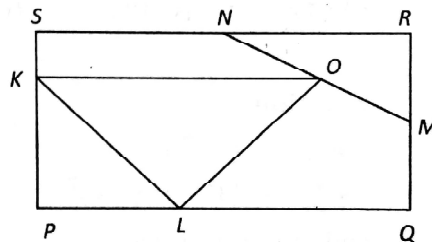


AMTI (NMTC) - 2011

KAPREKAR CONTEST - SUB-JUNIOR LEVEL

PART - A

1. Given $63.63 = m \left(21 + \frac{n}{100} \right)$, m, n positive integers with $n < 100$. The value of $(m + n)$ is
 (A) 21 (B) 24 (C) 104 (D) 101
2. n is a natural number and $(n + 2)(n + 4)$ is odd. Then the biggest power of 2 that divides $(n + 1)(n + 3)$ for any n is
 (A) 1 (B) 2 (C) 3 (D) 4
3. Two squares 17 cm x 17 cm overlap to form a rectangle 17 cm x 30 cm. The area of the overlapping region is
 (A) 289 (B) 68 (C) 510 (D) 85
4. Aruna and Baskar wrote the three digit number 888 and Aruna changed two of its digits and wrote the biggest three digit number divisible by 8. Bhaskar too changed two of the digits of 888 and wrote the smallest three digit number divisible by 8. The difference of the new numbers obtained is
 (A) 848 (B) 856 (C) 864 (D) 872
5. n is a negative integer. The expression having the greatest value is
 (A) $-2n^2 + 2n$ (B) $-2n^2 - 2n$ (C) $2n^2 + 2n$ (D) $2n^2 - 2n$
6. Five positive integers are written around a circle so that no two or three adjacent numbers have a sum divisible by 3. In this collection the number of numbers divisible by 3 is
 (A) 0 (B) 1 (C) 2 (D) 3
7. A number is called a palindrome if it reads the same forward or backward. For example 13531 is a palindrome. The difference between the biggest 10 digit palindrome and the smallest 9 digit palindrome is
 (A) 976666666 (B) 9888888888 (C) 9899999998 (D) 9777777777
8. In the figure, PQRS is a rectangle of area 2011 square units. K, L, M, N are the mid point of the respective sides. O is the midpoint of MN. The area of the triangle OKL is equal to (in square units)

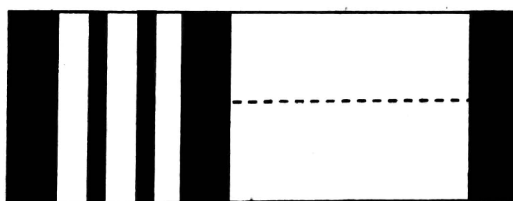


- (A) $\frac{2011}{5}$ (B) $\frac{2(2011)}{5}$ (C) $\frac{2011}{4}$ (D) $\frac{3(2011)}{8}$
9. A computer is printing a list of the seventh powers of all natural numbers, that is the sequence $1^7, 2^7, 3^7, \dots$. The number of terms (or numbers) between 5^{21} and 4^{28} are
 (A) 12 (B) 130 (C) 14 (D) 150
10. A natural number n has exactly two divisors and $(n + 1)$ has three divisors. The number of divisors of $(n + 2)$ is
 (A) 2 (B) 3
 (C) 4 (D) depends on the value of n
11. $\frac{p}{9} + \frac{q}{10} = r$ where p, q, r are positive integers and $p = q$. The biggest value of r less than 100 is
 (A) 19 (B) 57 (C) 76 (D) 95

12. In the grid shown in the diagram A and B erase 4 numbers each from the table. The sum of the numbers erased by them are in the ratio 10:11. After erasing the numbers, the number left out on the table is
- | | | |
|----|----|----|
| 13 | 14 | 15 |
| 20 | 21 | 22 |
| 27 | 28 | 29 |
- (A) 21 (B) 14 (C) 22 (D) 13
13. The positive integers p, q, $p - q$ and $p + q$ are all prime numbers. The sum of all these numbers is
- (A) divisible by 3 (B) divisible by 5
(C) divisible by 7 (D) prime
14. a, b, c are positive reals such that $a(b + c) = 32$, $b(c + a) = 65$ and $c(a + b) = 77$. Then $abc =$
- (A) 100 (B) 110 (C) 220 (D) 130
15. 100th term of the sequence 1, 3, 3, 3, 5, 5, 5, 5, 5, 7, 7, 7, 7, 7, 7, 7 is
- (A) 15 (B) 13 (C) 17 (D) 19

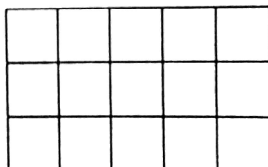
PART - B

1. When a barrel is 40 % empty it contains 80 litres more than when it is 20% full. The full capacity of the barrel (in litres) is _____.
2. a, b, c are the digits of a nine digit number abcabcabc. The quotient when this number is divided by 1001001 is _____
3. The first term of a sequence is $\left(\frac{2}{7}\right)$. Each new term is calculated using the formula $\left(\frac{1-x}{1+x}\right)$ where x is the preceding term. The sum of the first 2011 terms is _____.
4. The number of integers n for which $\frac{n}{20-n}$ is the square of an integer is _____.
5. All sides of the convex pentagon ABCDE are equal in length. $\angle A = \angle B = 90^\circ$. Then $\angle E$ is equal to _____.
6. The units digit of the number 3^{2011} is _____.
7. The number of digits used to write all page numbers in a book is 192. The total number of pages in the book is _____.
8. A bar code is formed using 25 black and certain white bars. White and black bars alternate. The first and the last are black bars. Some of the black bars are thin and others are wide.



The number of white bars is 15 more than the thin black bars. The number of thick black bars is _____.

9. By drawing 10 lines of which 4 are horizontal and 6 are vertical crossing each other as in the figure, one can get 15 cells. With the same 10 lines of which 3 are vertical and 7 horizontal we get 12 cells.



10. If $a^2 - b^2 = 2011$ where a, b are integers, then the most negative value of $(a + b)$ is _____.
11. x and y are real numbers such that $xy = x + y = \frac{x}{y}$ ($x \neq 0, y \neq 0$). Then the numerical value of $(x - y)$ is _____.

12. When $x = 2011$, the value of $\frac{1}{1 + \frac{1}{1 + \frac{1}{1 + \frac{1}{x}}}}$ is _____.

13. After the school final exams are over, all students in a class exchange their photographs. Each student gives his photograph to each of the remaining students and gets the photograph of all his friends. There are totally 870 exchanges of photos. The number of students in the class is _____.
14. The angles of a polygon are in the ratio 2:4:5:6:6:7. The difference between the greatest and least angle of the polygon is _____.
15. The perimeter of a right angled triangle is 132. The sum of the squares of all its sides is 6050. The sum of the legs of the triangle is _____.
